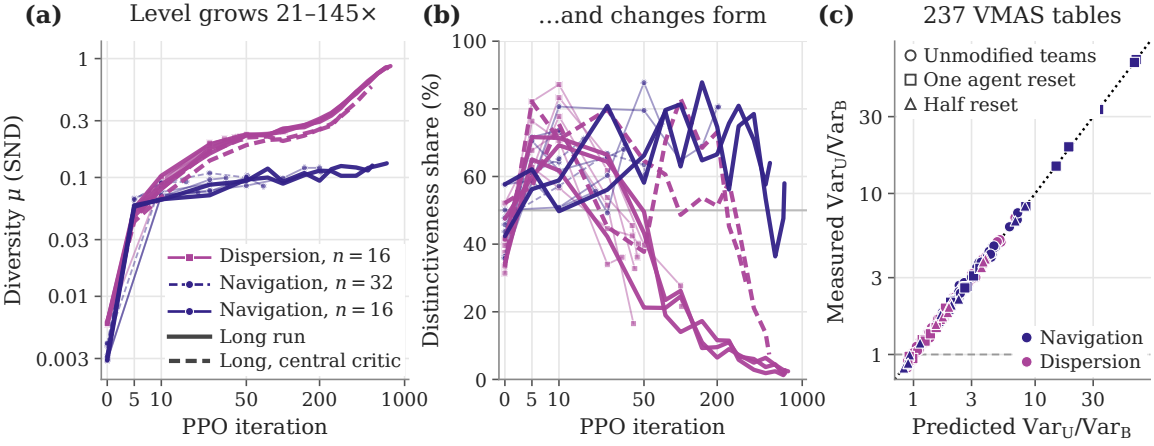
(a)
Level grows 21–145×
Diversity μ (SND)
1
0.3
0.1
0.03
0.01
0.003
Dispersion, n = 16
Navigation, n = 32
Navigation, n = 16
Long run
Long, central critic
0
5
10
50
200
1000
PPO iteration
(b)
...and changes form
Distinctiveness share (%)
100
80
60
40
20
0
0
5
10
50
200
1000
PPO iteration
(c)
237 VMAS tables
Unmodified teams
One agent reset
Half reset
Measured Var_U/Var_B
30
10
3
1
Navigation
Dispersion
1
3
10
30
Predicted Var_U/Var_B